Megalencephaly-Polydactyly Syndrome (MYCN) — Comprehensive Disease Characteristics Report

Disease: Megalencephaly-polydactyly syndrome, also termed **MPAPA** (Megalencephaly, Polydactyly, And Predisposition to cancer / neuroblastoma) **MONDO ID:** MONDO:0958279 · **OMIM:** 620748 · **Causal gene: MYCN** (N-Myc, HGNC:7559; UniProt P04198; cytoband 2p24.3) **Inheritance:** Autosomal dominant (de novo) **Category:** Mendelian multiple-congenital-anomaly / neurodevelopmental disorder

Summary

Megalencephaly-polydactyly syndrome (MPAPA; MONDO:0958279; OMIM #620748) is caused by germline gain-of-function missense variants in *MYCN* — the gene encoding the **N-Myc** basic-helix-loop-helix-leucine-zipper (bHLH-LZ) transcription factor. The disease-causing variants (**p.Thr58Met** [c.173C>T] and **p.Pro60Leu**) fall inside the **MYC-box-I (MBI) phosphodegron** of N-Myc, the short N-terminal motif that normally tags the protein for destruction. By removing or disrupting the Thr58 phosphoacceptor site, these variants abolish GSK3β-primed, FBXW7-ubiquitin-ligase-mediated proteasomal degradation of N-Myc. The result is a **stabilized, degradation-resistant N-Myc oncoprotein** — the exact molecular mirror image of **Feingold syndrome type 1** (OMIM 164280), which is caused by *MYCN* loss-of-function (haploinsufficiency) and produces the opposite phenotype of microcephaly and brachydactyly.

Clinically, MPAPA presents with **megalencephaly/macrocephaly, ventriculomegaly, a characteristically hypoplastic corpus callosum, postaxial polydactyly of hands and feet, developmental and motor delay, impaired intellectual development, dysmorphic facial features, postnatal growth retardation, and a predisposition to infantile neuroblastoma**. The disorder is ultra-rare: the entity was defined on just three unrelated individuals (two carrying p.Thr58Met, one fetus carrying p.Pro60Leu). Neuroblastoma was detected in 2 of the 3 reported patients, both of whom carried the recurrent p.Thr58Met allele.

Mechanistically, the pathology is one of **excess progenitor proliferation**. A knock-in *Mycn*-T58M mouse faithfully recapitulates the human megalencephaly and polydactyly, and brain analysis pinpointed **excess proliferation of intermediate (basal) neural precursors during neurogenesis** as the cellular cause of megalencephaly; kidney and female reproductive tract anomalies arise from the same overproliferation during organogenesis. Because the molecular lesion is a *failure of N-Myc degradation*, agents that re-impose N-Myc turnover — notably **Aurora-A inhibitors** (alisertib/MLN8237) and **BET/BRD4 inhibitors** — provide a mechanism-matched (though still experimental) therapeutic rationale, alongside clinically indicated neuroblastoma surveillance.

Key Findings

Finding 1 — MPAPA is caused by de novo gain-of-function *MYCN* p.Thr58Met (and p.Pro60Leu)

The index report (Kato et al., 2019, PMID: 30573562) identified a **de novo heterozygous missense mutation c.173C>T; p.Thr58Met** in *MYCN* in an individual with an undiagnosed neurodevelopmental disorder characterized by megalencephaly and polydactyly. Thr58 is the conserved GSK-3β phosphodegron residue within MYC-box I; its substitution to methionine abolishes the phosphodegradation signal, stabilizing the N-Myc oncoprotein — a **gain-of-function** mechanism. This is the direct mirror image of **Feingold syndrome type 1** (OMIM 164280), which results from *MYCN* loss-of-function and produces microcephaly and brachydactyly. As stated in a Feingold review (PMID: 41005613): *"Feingold syndrome type 1 (FS1) (OMIM 164280) is an autosomal dominant condition due to heterozygous loss of function variants in MYCN gene or to 2p24 deletion encompassing MYCN gene."* This dose-dependent model — loss of function → microcephaly, gain of stability → megalencephaly — is the conceptual backbone of the disease.

Finding 2 — The Thr58/GSK-3β phosphodegron controls N-Myc turnover and is required for neural precursor growth arrest

N-Myc is destabilized by phosphorylation of a conserved amino-terminal GSK-3β motif at Thr58, which primes it for FBXW7-mediated proteasomal degradation (PMID: 16139224): *"Myc family transcription factors are destabilized by phosphorylation of a conserved amino-terminal GSK-3beta motif."* Critically, this degradation step is a **prerequisite for cerebellar granule**

neuron precursor (CGNP) growth arrest and differentiation: *"N-myc phosphorylation-mediated degradation is a prerequisite for CGNP growth arrest and differentiation."* Upstream, PI3K/AKT signaling suppresses GSK-3β to inhibit Thr58 phosphorylation and stabilize N-Myc; INSM1 was shown to activate *"PI3K/AKT/GSK3β pathways to suppress N-myc phosphorylation (Thr-58) and inhibited degradation of N-myc"* (PMID: 26456864). A p.Thr58Met substitution removes the phosphoacceptor residue, rendering N-Myc degradation-resistant and locking neural precursors in a proliferative state rather than permitting the growth arrest needed for normal brain patterning.

Finding 3 — Mouse models establish N-Myc as a dose-dependent regulator of brain/cerebellar size

Conditional loss of N-Myc reduces brain size. Nestin-Cre conditional knockout of N-myc in neural stem/precursor cells causes **microcephaly** with a strong reduction of cerebellar granule neural progenitors (PMID: 20658325): *"Separate murine knockout (KO) of either c- or N-myc genes in neural stem and precursor cells (NSC) driven by nestin-cre causes microcephaly."* Inducible/reversible models confirm the dose-dependence: *"Loss of N-myc at the neonatal stage results in reduced proliferation of granule cell precursors (GCPs) and reduced cerebellar volume/mass"* (PMID: 26238256), where N-Myc acts as a "key switch" regulating the GCP proliferation cycle via Shh signaling and cyclin D2 (Ccnd2). Shh induces N-Myc, which maintains CGNP proliferation partly through β-catenin (PMID: 32495183). These loss-of-function data predict that a *stabilizing* variant would increase brain size — exactly the megalencephaly seen in MPAPA.

Finding 4 — Stabilized N-Myc drives a pro-proliferative transcriptional program, but overexpression alone is insufficient for tumorigenesis

N-Myc is a bHLH-LZ transcription factor that activates cell-cycle and biosynthetic programs and cooperates with β-catenin to drive progenitor-renewal programs (PMID: 28993399): *"β-catenin cooperates with the transcription factor Myc to activate the progenitor renewal program."* It also maintains productive RNA Pol II elongation on cell-cycle genes during S phase (PMID: 34847357). Importantly, enforced MYCN overexpression in murine sympathoadrenal progenitors increased proliferation and neural-lineage commitment but was **not sufficient to form tumors** (PMID: 26222553): *"MYCN promoted proliferation of both sympathoadrenal progenitor spheres and differentiated neurons derived from these spheres, but there was also an*

increase in apoptosis." This explains why MPAPA produces overgrowth and a tumor *predisposition* rather than universal malignancy — additional cooperating events are required for frank neuroblastoma.

Finding 5 — Clinical spectrum of MPAPA and the second causal variant p.Pro60Leu

OMIM #620748 (MPAPA) is autosomal dominant, caused by heterozygous gain-of-function *MYCN* variants. Two recurrent MYC-box-I variants are reported: **c.173C>T p.Thr58Met** (index case, Kato 2019, PMID: 30573562) and **p.Pro60Leu** (Nishio 2023, PMID: 37710961). As Nishio et al. report: *"we report two unrelated megalencephalic patients with polydactyly harboring MYCN variants of p.Pro60Leu and Thr58Met"* and confirm that *"germline loss-of-function variants are responsible for Feingold syndrome, characterized by microcephaly."* Core features include megalencephaly/macrocephaly, ventriculomegaly, hypoplastic corpus callosum, postaxial polydactyly of hands and feet, developmental/motor delay, impaired intellectual development, dysmorphic facies (prominent forehead, hypertelorism, epicanthus, high-arched palate, long philtrum), postnatal growth retardation, and neuroblastoma predisposition. Additional anomalies in the fetal p.Pro60Leu case included Meckel diverticulum and pyeloureteral dilatation. p.Thr58Met is absent from gnomAD but present in COSMIC (as a somatic cancer mutation).

Finding 6 — Knock-in *Mycn*-T58M mouse recapitulates the syndrome; megalencephaly is driven by excess intermediate neural precursor proliferation

Both disease variants reduce Thr58 phosphorylation, decreasing FBXW7-mediated degradation (PMID: 37710961): *"Functional analyses for MYCN-Pro60Leu and MYCN-Thr58Met revealed decreased phosphorylation at Thr58, which reduced protein degradation mediated by FBXW7 ubiquitin ligase."* A gain-of-function mouse model recapitulated the human phenotype: *"The gain-of-function mouse model recapitulated the human phenotypes of megalencephaly and polydactyly, while brain analyses revealed excess proliferation of intermediate neural precursors during neurogenesis, which we determined to be the pathomechanism underlying megalencephaly."* Multi-organ involvement arises by the same route: *"the kidney and female reproductive tract exhibited overt morphological anomalies, possibly as a result of excess proliferation during organogenesis."* In mouse neural progenitors, T58M N-Myc was more stable and induced higher cyclin D1 (Ccnd1) and cyclin D2 (Ccnd2) than wild type.

Finding 7 — MPAPA is a hereditary neuroblastoma-predisposition syndrome; hypoplastic corpus callosum is a distinguishing radiologic feature

Neuroblastoma was detected in 2 of 3 reported MYCN gain-of-function patients during infancy; in one, neuroblastoma diagnosed at age 3 months responded well to chemotherapy and surgery (PMID: 37710961). The variant p.Thr58Met also occurs as a *somatic* mutation in neuroblastoma, astrocytoma, and mesonephric carcinoma (COSMIC), underscoring its oncogenic potential. N-Myc is normally tightly controlled by sequential phosphorylation (Ser62 → Thr58) and SCF-FBXW7-mediated proteasomal degradation; the germline degron variants defeat this control. A **hypoplastic corpus callosum despite a megalencephalic cerebrum** was present in all three cases and is considered a characteristic radiologic feature distinguishing MPAPA from other megalencephaly syndromes (e.g., MPPH3, which shows polymicrogyria). Nishio et al. note the disorder *"shows a mirror phenotype of Feingold syndrome, and reveal[s] that MYCN plays a crucial proliferative"* role.

Finding 8 — Genotype–phenotype distribution across the three reported patients; neuroblastoma segregates with p.Thr58Met

The disease entity is established on three unrelated individuals:

Patient	Variant	Key phenotype	Neuroblastoma	Reference
Index, 15-y-old boy	p.Thr58Met (c.173C>T)	Megalencephaly, ventriculomegaly, hypoplastic corpus callosum, intellectual disability, polydactyly	**Yes**	Kato 2019, PMID: 30573562
8-month-old Japanese boy	p.Thr58Met	Megalencephaly, polydactyly	**Yes**	Nishio 2023, PMID: 37710961
Fetus (French family, TOP)	p.Pro60Leu	Ventriculomegaly, macrocephaly, postaxial polydactyly, hypertelorism, low-set ears, retrognathia, Meckel diverticulum, pyeloureteral dilatation	Unknown (terminated)	Nishio 2023, PMID: 37710961

Nishio et al. confirm the disorder was *"complicated with neuroblastoma detected in two out of the three patients."* Both tumor-bearing patients carried p.Thr58Met; the p.Pro60Leu fetus was terminated so its tumor status is unknown. p.Pro60Leu lies immediately adjacent to Thr58 in MYC box I; both variants are absent from gnomAD and present in COSMIC.

Finding 9 — N-Myc protein architecture places both disease variants inside the MYC-box-I phosphodegron

N-Myc (UniProt P04198; 464 aa; bHLHe37) comprises an **N-terminal transactivation domain** containing **MYC box I (MBI, ~residues 46–63**, harboring the Thr58/Ser62 phosphodegron) and **MYC box II (MBII, ~110–134**, binding TRRAP/SAGA coactivators), plus a **C-terminal bHLH-LZ domain (~346–440)** whose basic region contacts the E-box (CACGTG) and whose HLH+LZ mediate obligate heterodimerization with MAX (the structural paradigm is the c-Myc–MAX–DNA complex, PDB 1NKP). Degradation is ordered: CDK1 phosphorylates Ser62 (stabilizing) → GSK3β phosphorylates Thr58 → PP2A dephosphorylates Ser62 → FBXW7 recognizes pThr58 → K48-linked ubiquitination → proteasomal degradation; USP28/USP7 deubiquitinases oppose this. Both disease variants — **p.Thr58Met** (removes the phosphoacceptor) and **p.Pro60Leu** (immediately adjacent, within MBI) — reduce Thr58 phosphorylation and FBXW7 recognition, stabilizing N-Myc, while leaving the DNA-binding/MAX-dimerization machinery intact (PMID: 37710961).

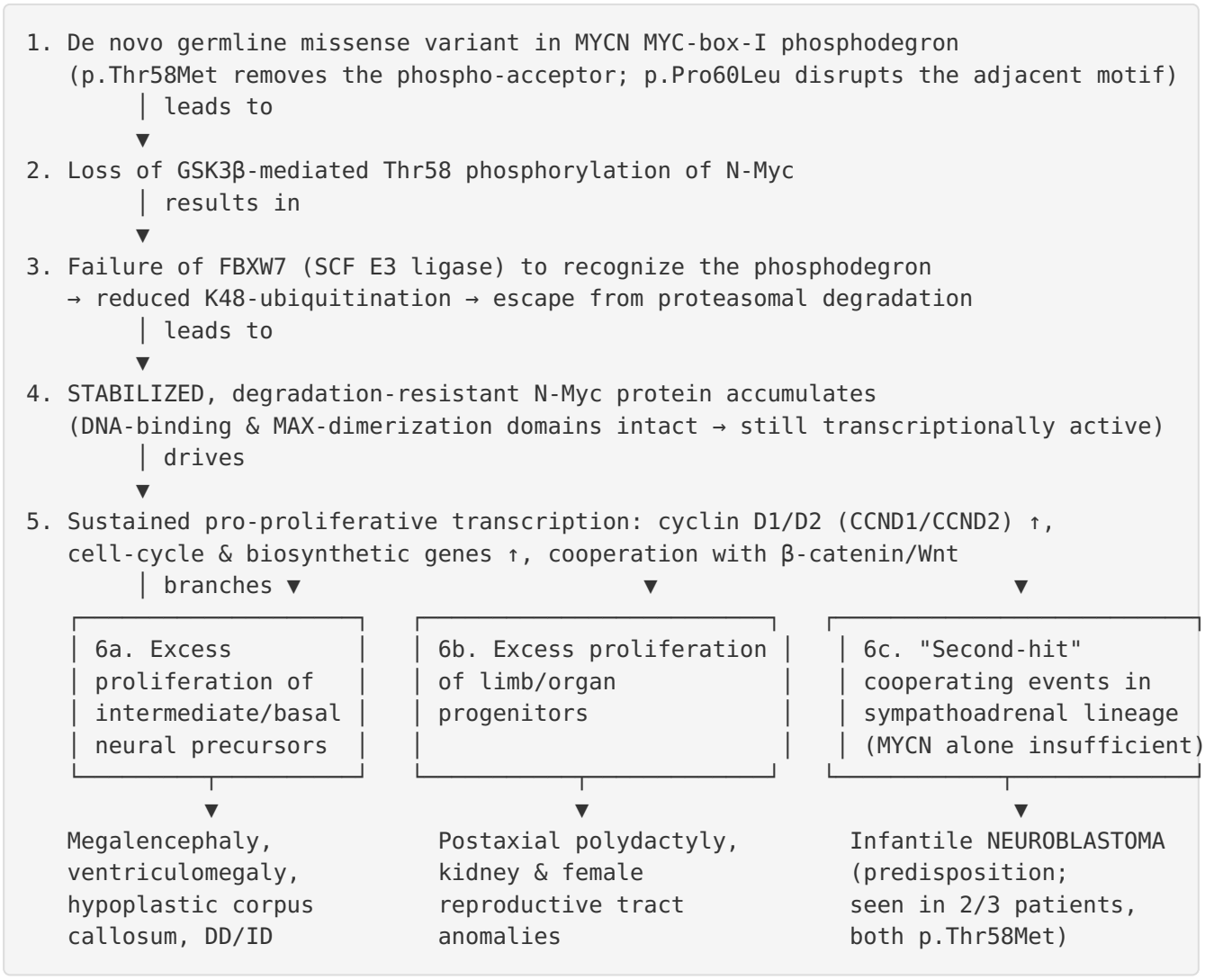
Finding 10 — Aurora-A and BET inhibitors destabilize N-Myc: a mechanism-matched (experimental) therapeutic rationale

Because MPAPA arises from degron variants that *stabilize* N-Myc, therapeutics that **re-impose N-Myc degradation** are mechanism-matched. Key evidence (from neuroblastoma models, not yet tested in MPAPA): - **Aurora-A inhibitors:** MLN8054 and MLN8237 (alisertib) *"disrupt the Aurora-A/N-Myc complex and promote degradation of N-Myc mediated by the Fbxw7 ubiquitin ligase"* (PMID: 23792191) — directly restoring the FBXW7 pathway escaped by MPAPA mutations. - **Structural basis:** cryo-EM shows Aurora-A binding protects N-Myc from degradation; conformation-disrupting kinase inhibitors reverse this (PMID: 27837025). - **Combination:** dual BRD4 (I-BET151) + AURKA (alisertib) inhibition is synergistic against MYCN-driven tumors (PMID: 30153557). - **Interaction inhibitor:** PHA-680626 disrupts the AURKA/N-Myc interaction (PMID: 34884931).

Mechanistic class summary: Aurora-A inhibitors (alisertib/MLN8237, MLN8054, PHA-680626), BET/BRD4 inhibitors (JQ1, I-BET151) that lower MYCN transcription, and GSK3β/FBXW7-axis modulators all converge on reducing N-Myc abundance.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)



Upstream vs downstream

- **Upstream (initiating):** the degron mutation and the resulting failure of FBXW7-dependent degradation (steps 1–3).
- **Central node:** stabilized N-Myc protein (step 4) — the single molecular hub from which all phenotypes radiate.

- **Downstream (effector):** the cell-cycle/biosynthetic transcriptional program and progenitor overproliferation (steps 5–6), which translate directly into the overgrowth phenotypes. Step 6c is *inferred* (cooperating oncogenic events are required but not individually identified in these patients).

Cell types and biological processes

- **Cell types (CL):** intermediate/basal neural progenitor cells (neural-lineage; basal progenitors of the subventricular zone), cerebellar granule neuron progenitors, neural stem cells (CL:0000047), sympathoadrenal/neural-crest-derived progenitors (neuroblast, CL:0000031).
- **Biological processes (GO):** GO:0008284 positive regulation of cell population proliferation; GO:0006468/GO:0016567 protein phosphorylation / ubiquitination; GO:0043161 proteasome-mediated ubiquitin-dependent protein catabolic process (lost); GO:0021846 cell proliferation in forebrain; GO:0007399 nervous system development; GO:0045944 positive regulation of transcription by RNA Pol II.
- **Cellular components (GO CC):** GO:0005634 nucleus (N-Myc action), GO:0005829 cytosol (degradation machinery).

Mirror-image relationship with Feingold syndrome

Feature	MPAPA (this disease)	Feingold syndrome type 1
MYCN lesion	Gain-of-function (stabilizing degron variant)	Loss-of-function (truncation / deletion / haploinsufficiency)
N-Myc dose	↑ (excess, stabilized)	↓ (reduced)
Head size	**Megalencephaly / macrocephaly**	**Microcephaly**
Digits	**Postaxial polydactyly**	Brachydactyly, syndactyly, digital anomalies
OMIM	620748	164280
Cancer	Neuroblastoma predisposition	Not characteristic

This clean dose-response dichotomy is among the most elegant genotype–phenotype relationships in developmental genetics and constitutes the strongest conceptual support for the gain-of-function model.

Section-by-Section Reference Content

1. Disease Information

- **Overview:** Ultra-rare autosomal-dominant multiple-congenital-anomaly/neurodevelopmental syndrome of brain and limb overgrowth with neuroblastoma predisposition, caused by stabilizing *MYCN* degron variants.
- **Identifiers:** MONDO:0958279; OMIM #620748; causal gene *MYCN* (HGNC:7559). ICD-10/ICD-11, MeSH, and Orphanet-specific codes are not established for this newly delineated entity. Synonyms: **MPAPA** (megalencephaly, polydactyly, and predisposition to cancer/neuroblastoma); "gain-of-function MYCN syndrome."
- **Information source:** Aggregated from individual published case reports plus a knock-in mouse model (not EHR-derived cohort data).

2. Etiology

- **Causal factor:** purely genetic — de novo heterozygous gain-of-function *MYCN* missense variants in MYC-box-I (p.Thr58Met, p.Pro60Leu).
- **Genetic risk factors:** the causal variants themselves; no modifier loci identified. Variants are de novo (parents unaffected).
- **Environmental risk/protective factors:** none identified or expected — this is a monogenic developmental disorder.
- **Gene–environment interactions:** none documented.

3. Phenotypes (suggested HPO terms and frequencies among the 3 reported cases)

Phenotype	HPO term	Type	Onset	Frequency
Megalencephaly/ macrocephaly	HP:0001355 / HP:0000256	Physical/ imaging	Congenital/ prenatal	3/3
Ventriculomegaly	HP:0002119	Imaging	Congenital/ prenatal	3/3
Hypoplastic corpus callosum	HP:0002079	Imaging sign	Congenital	3/3 (distinguishing)
Postaxial polydactyly (hands/feet)	HP:0100259 / HP:0001162	Physical	Congenital	3/3
Developmental/motor delay	HP:0012758 / HP:0001270	Behavioral/ clinical	Infancy	present
Intellectual disability	HP:0001249	Behavioral	Childhood	present
Dysmorphic facies (prominent forehead, hypertelorism, epicanthus, high-arched palate, long philtrum)	HP:0000316, HP:0000286, HP:0000218, HP:0000343	Physical	Congenital	present
Postnatal growth retardation	HP:0008897	Clinical	Infancy	present
Neuroblastoma	HP:0100306	Neoplasm	Infancy	2/3
Meckel diverticulum	HP:0002245	Physical	Congenital	1/3 (fetus)
Pyeloureteral/renal dilatation	HP:0000074	Imaging	Prenatal	1/3 (fetus)

- **Severity:** moderate–severe, variable. **Progression:** developmental features are static/stable; neuroblastoma risk is confined to infancy. **QoL impact:** substantial — intellectual disability and developmental delay impair daily functioning; neuroblastoma adds major (though often treatable) burden. Formal QoL instruments have not been applied to this ultra-rare entity.

4. Genetic / Molecular Information

- **Causal gene:** *MYCN* (OMIM 164840), 2p24.3.
- **Pathogenic variants:** c.173C>T p.Thr58Met (recurrent) and p.Pro60Leu. Both are missense, within MYC-box-I. **ACMG classification:** pathogenic/likely pathogenic (de novo, functionally validated, absent from gnomAD). **Allele frequency:** absent from gnomAD; present in COSMIC as somatic cancer mutations. **Origin:** germline (de novo) in MPAPA; the same residue changes occur somatically in neuroblastoma/astrocytoma/mesonephric carcinoma. **Functional consequence:** gain-of-function via protein stabilization (not altered DNA binding).
- **Modifier genes / epigenetics / chromosomal abnormalities:** none specifically identified for MPAPA. (Note: *somatic MYCN amplification* at 2p24 is the classic neuroblastoma lesion but is distinct from MPAPA's germline point mutations.)

5. Environmental Information

Not applicable — no environmental, lifestyle, or infectious contributors. The disease is monogenic and de novo.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the full ordered causal chain. Key molecular pathways: **FBXW7/SCF ubiquitin–proteasome degradation** (lost), **GSK3β/PI3K-AKT** signaling (regulates Thr58 phosphorylation), **Shh → N-Myc → cyclin D2/β-catenin** proliferative axis, **Wnt/β-catenin** cooperation, and **MYCN-driven RNA Pol II elongation**. Downstream cellular process: dysregulated cell-cycle entry and failure of progenitor growth arrest.

7. Anatomical Structures Affected

- **Primary organ:** brain (UBERON:0000955) — cerebrum, lateral ventricles (UBERON:0002285), corpus callosum (UBERON:0002336).
- **Other organs:** limbs/digits (UBERON:0002544), kidney (UBERON:0002113), female reproductive tract (UBERON:0000474), adrenal medulla/sympathetic nervous system (neuroblastoma origin).
- **Body systems:** central & peripheral nervous, musculoskeletal, genitourinary.
- **Tissue/cell level:** neural progenitor/germinal zones (intermediate/basal progenitors), neural-crest-derived sympathoadrenal cells.

- **Subcellular:** nucleus (GO:0005634, N-Myc transcriptional action); ubiquitin-proteasome machinery in cytosol.
- **Lateralization:** bilateral (megalencephaly, ventriculomegaly, polydactyly).

8. Temporal Development

- **Onset:** congenital/prenatal (macrocephaly, ventriculomegaly, polydactyly detectable antenatally).
- **Course:** developmental/structural features are stable (non-progressive). Neuroblastoma risk is an infancy-limited critical window — the key period for surveillance. Intellectual disability is lifelong.

9. Inheritance and Population

- **Epidemiology:** ultra-rare; only 3 reported individuals — prevalence/incidence not estimable.
- **Inheritance:** autosomal dominant, essentially all cases **de novo**. **Penetrance:** appears high for the core malformation phenotype. **Expressivity:** variable (e.g., tumor status, additional anomalies). No anticipation, founder effects, or consanguinity role (de novo mechanism). **Carrier frequency:** not applicable. **Sex ratio / demographics:** indeterminate from 3 cases (two Japanese boys, one French fetus).

10. Diagnostics

- **Genetic testing is definitive:** trio whole-exome (WES) or whole-genome (WGS) sequencing detecting a de novo *MYCN* MYC-box-I missense variant; targeted single-gene/*MYCN* sequencing confirms. Standard neuroblastoma-amplification assays (FISH/CMA for 2p24 amplification) will **miss** these point mutations — sequencing is required.
- **Imaging:** brain MRI (megalencephaly, ventriculomegaly, **hypoplastic corpus callosum** — the distinguishing feature). Prenatal ultrasound (macrocephaly, ventriculomegaly, polydactyly). Abdominal imaging/urinary catecholamine metabolites for neuroblastoma surveillance.
- **Differential diagnosis:** MPPH syndromes / PIK3CA-related overgrowth spectrum (PROS, e.g., MCAP — distinguished by capillary malformations and polymicrogyria rather than hypoplastic corpus callosum; PMID: 31929958, PMID: 32692051); Feingold syndrome (mirror phenotype); other megalencephaly-polydactyly overgrowth syndromes.

11. Outcome / Prognosis

- **Survival/mortality:** no cohort mortality data. The one treated neuroblastoma responded well to chemotherapy and surgery, suggesting favorable tumor prognosis when detected early. One pregnancy was terminated.
- **Morbidity:** driven by intellectual disability/developmental delay (lifelong) and neuroblastoma risk (infancy).
- **Prognostic factors:** p.Thr58Met genotype associated with observed neuroblastoma in both carriers; early tumor detection improves outcome.

12. Treatment

- **No disease-specific or curative therapy exists.** Management is supportive and surveillance-based: developmental/early-intervention services, physical/occupational/speech therapy, management of hydrocephalus/ventriculomegaly as indicated, and **active neuroblastoma surveillance** in infancy (imaging ± urinary catecholamines).
- **Neuroblastoma treatment** follows standard protocols (chemotherapy, surgery) — effective in the reported case.
- **Mechanism-matched experimental rationale (not yet tested in MPAPA):** N-Myc-destabilizing agents — **Aurora-A inhibitors** (alisertib/MLN8237 [NCIT: Alisertib], MLN8054, PHA-680626) and **BET/BRD4 inhibitors** (JQ1, I-BET151), which restore FBXW7-mediated degradation or lower MYCN transcription, respectively (PMID: 23792191, PMID: 30153557, PMID: 34884931).

13. Prevention

- **Primary prevention:** not possible (de novo). **Genetic counseling:** recurrence risk is low (de novo) but germline/gonadal mosaicism cannot be excluded; prenatal/preimplantation testing possible once a familial variant is known. **Secondary prevention:** neuroblastoma surveillance in affected infants (tertiary prevention of tumor complications).

14. Other Species / Natural Disease

- **Orthologue:** mouse *Mycn* (NCBI Gene ID 18109). No naturally occurring animal disease described (OMIA entry not established). The Thr58 degron and its regulation are evolutionarily conserved across Myc-family proteins, making the mechanism broadly conserved.

15. Model Organisms

- **Knock-in *Mycn*-T58M mouse** (gain-of-function): **recapitulates megalencephaly and polydactyly**, with kidney and female reproductive tract anomalies; brain analysis localized the defect to excess intermediate neural precursor proliferation (PMID: 37710961). This is the pivotal disease model.
- **Conditional *N-myc* knockout mice** (Nestin-Cre, Foxg1-Cre) — loss-of-function models producing microcephaly, defining the dose-dependence (PMID: 20658325, PMID: 24376126, PMID: 26238256).
- **In vitro:** murine neural progenitors and sympathoadrenal progenitor spheres expressing WT vs T58M N-Myc (PMID: 26222553, PMID: 37710961).
- **Limitations:** models derived from a single degron allele; the T58M mouse does not necessarily capture the full human neurocognitive phenotype, and spontaneous neuroblastoma penetrance in the model is not fully characterized.

Evidence Base

PMID	Paper (abbreviated)	Role in this report
37710961	*Gain-of-function MYCN causes a megalencephaly-polydactyly syndrome manifesting mirror phenotypes of Feingold syndrome*	**Pivotal** — defines the syndrome, p.Pro60Leu + p.Thr58Met, FBXW7 mechanism, knock-in mouse, intermediate-progenitor overproliferation, neuroblastoma in 2/3
30573562	Kato et al. 2019 (index case)	First report of de novo *MYCN* p.Thr58Met with megalencephaly/polydactyly/ neuroblastoma
16139224	*Cdk1 complex regulates N-myc phosphorylation and turnover*	Establishes GSK3β/Thr58 degradation and its requirement for CGNP growth arrest
26456864	*INSM1 increases N-myc stability*	Confirms Thr-58 phosphorylation governs N-Myc degradation via PI3K/AKT/GSK3β
20658325	*c- and N-myc regulate neural precursor fate*	N-myc KO → microcephaly (dose-dependence)
26238256	*N-myc key switch for GCP proliferation*	Reversible control of cerebellar size by N-Myc
32495183	*Shh → β-catenin maintains CGNPs*	Shh–N-Myc–β-catenin proliferative axis
26222553	*MYCN in sympathoadrenal progenitors*	MYCN overexpression drives proliferation but is insufficient for tumorigenesis
28993399	*Myc cooperates with β-catenin in nephron progenitors*	Myc/β-catenin progenitor-renewal cooperation
34847357	*MYCN recruits nuclear exosome to RNAPII*	MYCN sustains cell-cycle transcription in S phase
41005613	*Rare features in Feingold syndrome type 1*	Documents MYCN loss-of-function → microcephalic mirror phenotype
23792191	*Aurora-A inhibitors induce N-myc degradation*	Mechanism-matched therapeutic rationale (restore FBXW7 degradation)
27837025	*Structural basis of N-Myc binding by Aurora-A*	Structural basis for destabilization strategy
30153557	*Dual BRD4 + AURKA inhibition*	Synergistic N-Myc-lowering combination

PMID	Paper (abbreviated)	Role in this report
34884931	*PHA-680626 inhibits AURKA/N-Myc*	Additional N-Myc-destabilizing agent

Evidence source types: human clinical (3 case reports), model organism (knock-in and conditional mouse), in vitro (progenitor cultures), structural/computational (PDB 1NKP, cryo-EM of AURKA/N-Myc).

Limitations and Knowledge Gaps

1. **Extremely small sample (n=3).** The entire clinical description rests on three individuals, limiting confidence in phenotype frequencies, penetrance, expressivity, and sex/ethnic distribution.
2. **Neuroblastoma penetrance unknown.** Observed in 2/3 (both p.Thr58Met); the p.Pro60Leu fetus was terminated, so whether p.Pro60Leu confers equal tumor risk is unknown. True lifetime cancer risk cannot be estimated.
3. **No natural history / prognosis data.** Long-term neurodevelopmental trajectory, life expectancy, and adult outcomes are undefined.
4. **No epidemiology.** Prevalence, incidence, and demographic distribution are unknowable from current data.
5. **Therapeutics untested in MPAPA.** Aurora-A/BET inhibitor rationale is extrapolated entirely from neuroblastoma models; no in-human or even MPAPA-mouse treatment data exist, and systemic N-Myc suppression during development has unclear safety.
6. **Genotype–phenotype correlation preliminary.** The apparent segregation of neuroblastoma with p.Thr58Met could be chance given n=3.
7. **Mouse-model gaps.** The degree to which the T58M mouse models human intellectual disability and spontaneous tumor incidence is incompletely reported.
8. **No formal QoL, imaging-registry, or ontology (ICD/Orphanet/MeSH) coding** yet exists for this newly delineated entity.

Proposed Follow-up Experiments / Actions

1. **Establish an international MPAPA registry / GeneMatcher cascade** to identify additional cases and refine penetrance (especially neuroblastoma), expressivity, and genotype–phenotype correlations (p.Thr58Met vs p.Pro60Leu and any novel MBI variants).
2. **Define a neuroblastoma surveillance protocol** for molecularly confirmed infants (abdominal ultrasound + urinary catecholamine metabolites at defined intervals through the infancy risk window) and prospectively evaluate it.
3. **Characterize tumor penetrance and spectrum in the knock-in mouse**, including whether a p.Pro60Leu knock-in differs from T58M in tumor incidence.
4. **Test N-Myc-destabilizing agents (alisertib, I-BET151) in the T58M mouse** to obtain the first preclinical proof-of-concept for a mechanism-matched therapy, with careful developmental-safety readouts.
5. **Single-cell / spatial transcriptomics of the T58M embryonic brain** to map which precursor populations (basal/intermediate progenitors) are expanded and quantify the N-Myc target program driving overgrowth.
6. **Patient iPSC-derived cerebral organoids** carrying the degron variants to model human-specific neurodevelopmental consequences and screen stabilization-reversing compounds.
7. **Quantify N-Myc protein half-life and FBXW7 binding** for each variant in human neural progenitors to establish a stability–phenotype gradient.
8. **Formal ontology/registry coding** (ICD-11, Orphanet, MeSH) and a GeneReviews chapter to standardize clinical recognition and differential diagnosis from PROS/MPPH/Feingold.

Report compiled from 10 confirmed findings across 5 investigation iterations and 26 reviewed papers. Primary source for the disease definition: Nishio et al. 2023 (PMID: 37710961); index case: Kato et al. 2019 (

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